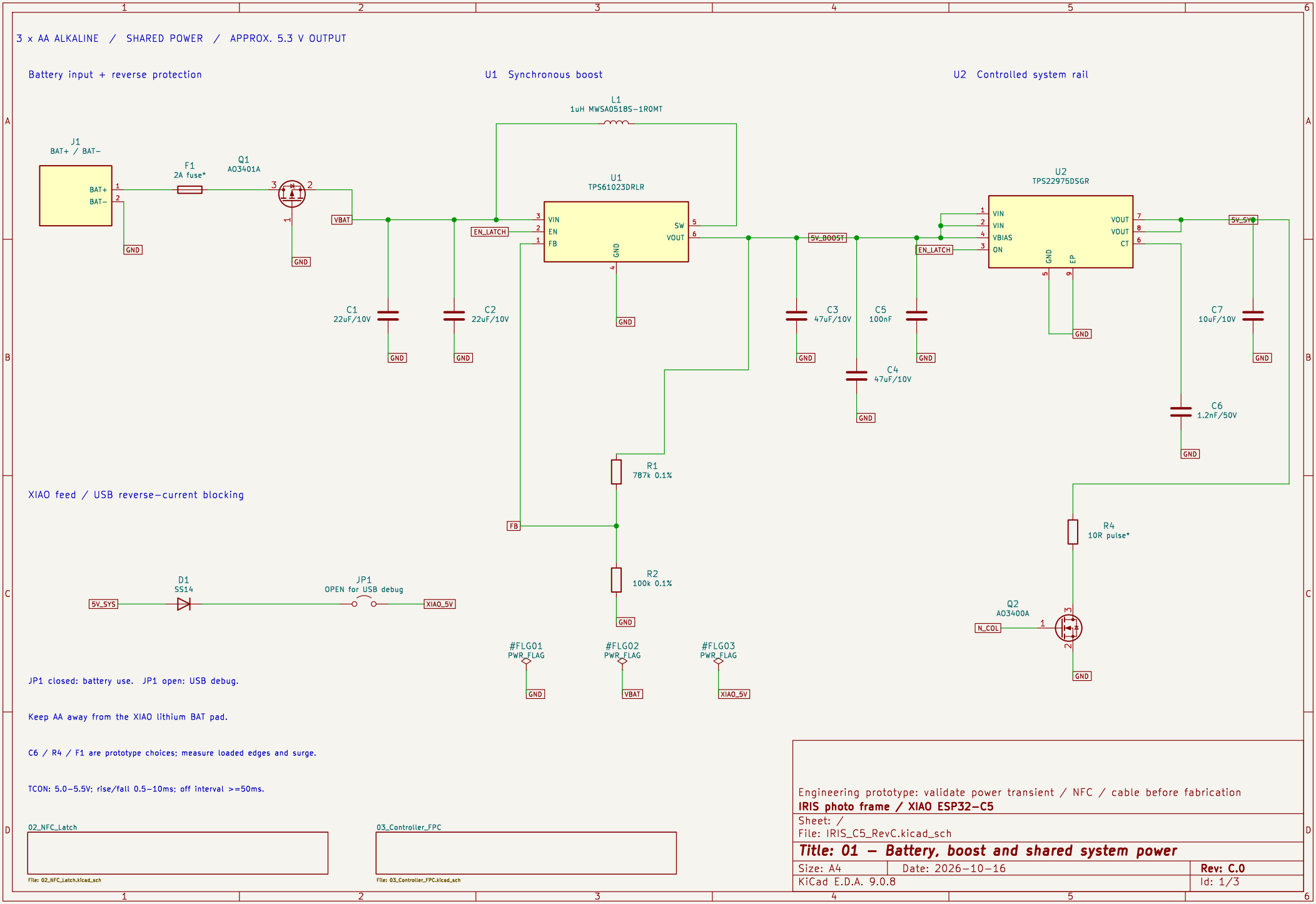
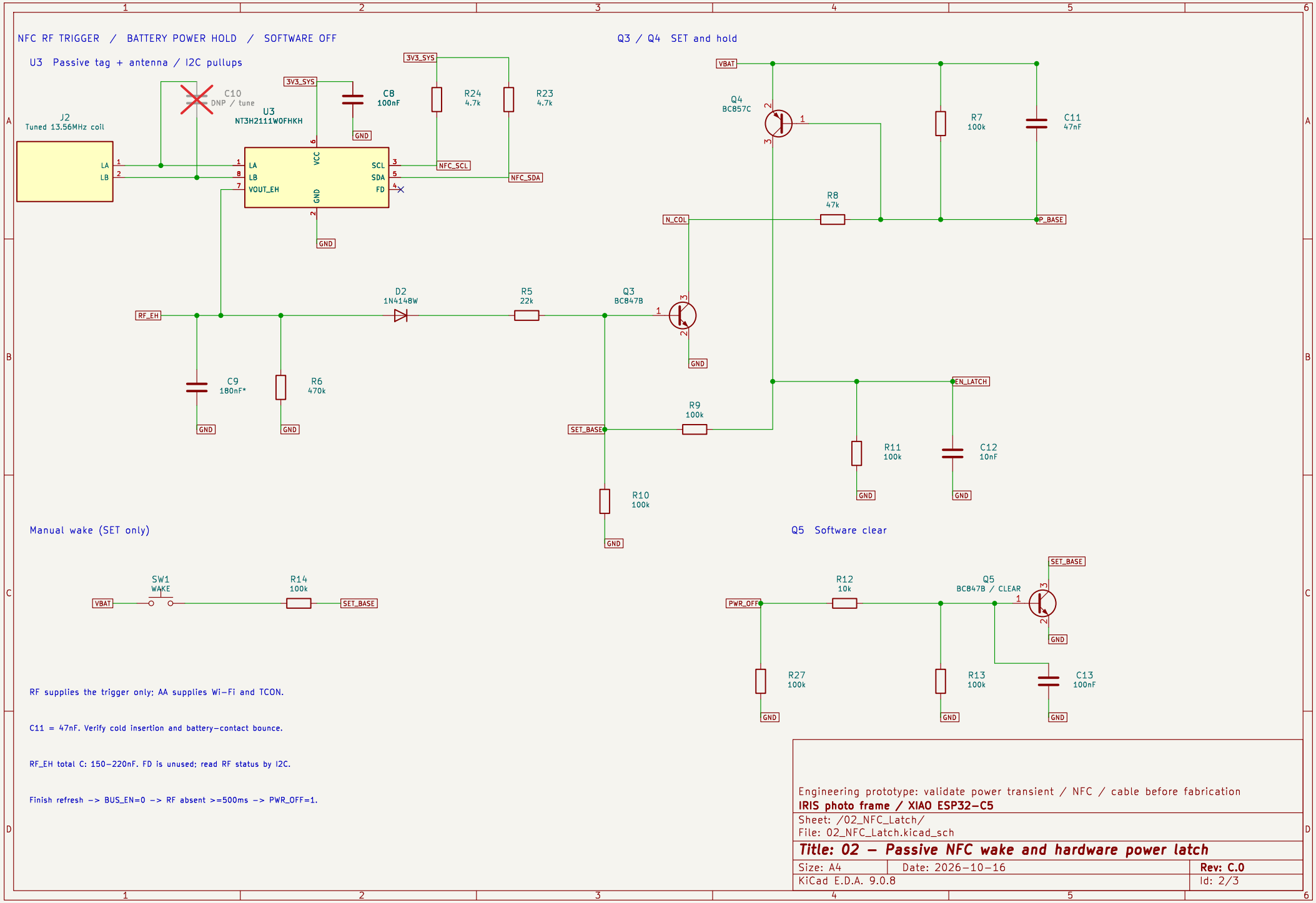
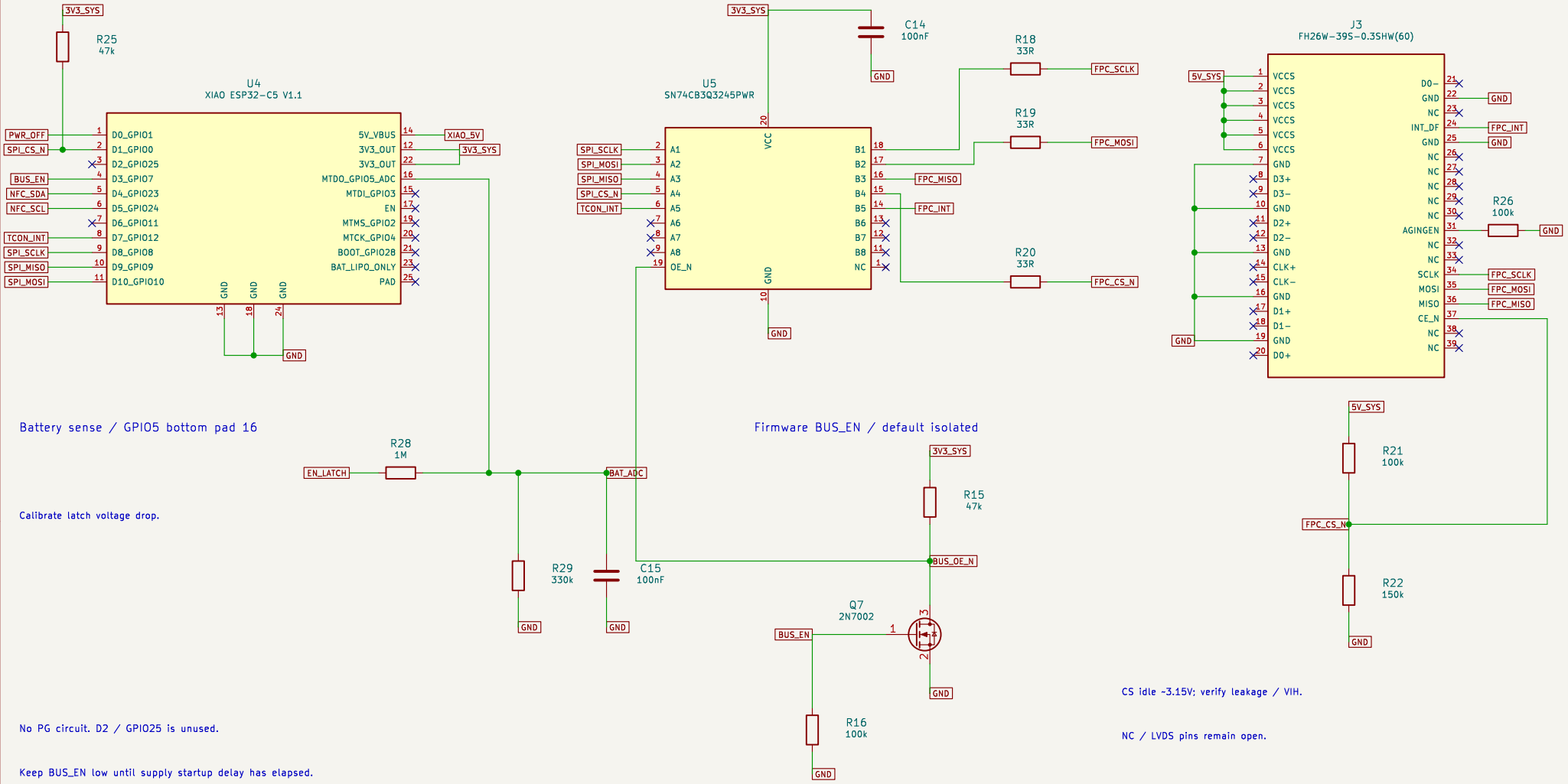

3 x AA ALKALINE / SHARED POWER / APPROX. 5.3 V OUTPUT
Battery input + reverse protection
U1 Synchronous boost
U2 Controlled system rail
J1
BAT+ / BAT−
BAT+
BAT−
F1
2A fuse*
Q1
AO3401A
GND
VBAT
L1
1uH MWSA0518S-1R0MT
U1
TPS61023DRLR
VIN
EN
FB
SW
VOUT
EN_LATCH
5V_BOOST
U2
TPS22975DSGR
VIN
VIN
VBIAS
ON
GND
EP
VOUT
VOUT
CT
5V_SYS
C1
22uF/10V
C2
22uF/10V
C3
47uF/10V
C4
47uF/10V
C5
100nF
C6
1.2nF/50V
C7
10uF/10V
R1
787k 0.1%
R2
100k 0.1%
R4
10R pulse*
XIAO feed / USB reverse-current blocking
D1
SS14
JP1
OPEN for USB debug
XIAO_5V
Q2
AO3400A
N_COL
#FLG01
PWR_FLAG
#FLG02
PWR_FLAG
#FLG03
PWR_FLAG
JP1 closed: battery use. JP1 open: USB debug.
Keep AA away from the XIAO lithium BAT pad.
C6 / R4 / F1 are prototype choices; measure loaded edges and surge.
TCON: 5.0–5.5V; rise/fall 0.5–10ms; off interval >=50ms.
02_NFC_Latch
File: 02_NFC_Latch.kicad_sch
03_Controller_FPC
File: 03_Controller_FPC.kicad_sch
Engineering prototype: validate power transient / NFC / cable before fabrication
IRIS photo frame / XIAO ESP32-C5
Sheet: /
File: IRIS_C5_RevC.kicad_sch
Title: 01 – Battery, boost and shared system power
Size: A4
Date: 2026-10-16
Rev: C.0
KiCad E.D.A. 9.0.8
Id: 1/3


NFC RF TRIGGER / BATTERY POWER HOLD / SOFTWARE OFF
U3 Passive tag + antenna / I2C pullups
J2
Tuned 13.56MHz coil
LA
LB
C10
DNP / tune
U3
NT3H2111W0FHKH
3V3_SYS
C8
100nF
GND
R24
4.7k
R23
4.7k
NFC_SCL
NFC_SDA
VCC
SCL
SDA
FD
VOUT_EH
RF_EH
D2
1N4148W
R5
22k
Q3
BC847B
C9
180nF*
R6
470k
R9
100k
R10
100k
SET_BASE
Q3 / Q4 SET and hold
VBAT
Q4
BC857C
R7
100k
C11
47nF
R8
47k
N_COL
P_BASE
EN_LATCH
R11
100k
C12
10nF
Manual wake (SET only)
SW1
WAKE
R14
100k
Q5 Software clear
R12
10k
Q5
BC847B / CLEAR
PWR_OFF
R27
100k
R13
100k
C13
100nF
RF supplies the trigger only; AA supplies Wi-Fi and TCON.
C11 = 47nF. Verify cold insertion and battery-contact bounce.
RF_EH total C: 150–220nF. FD is unused; read RF status by I2C.
Finish refresh -> BUS_EN=0 -> RF absent >=500ms -> PWR_OFF=1.
Engineering prototype: validate power transient / NFC / cable before fabrication
IRIS photo frame / XIAO ESP32-C5
Sheet: /02_NFC_Latch/
File: 02_NFC_Latch.kicad_sch
Title: 02 – Passive NFC wake and hardware power latch
Size: A4
Date: 2026-10-16
Rev: C.0
KiCad E.D.A. 9.0.8
Id: 2/3

XIAO ESP32-C5 / 3.3 V SPI / IRIS 39-PIN FPC



Engineering prototype: validate power transient / NFC / cable before fabrication

IRIS photo frame / XIAO ESP32-C5

Sheet: /03_Controller_FPC/
File: 03_Controller_FPC.kicad_sch

Title: 03 – XIAO controller, SPI isolation and FPC

Size: A4	Date: 2026-10-16	**Rev: C.0**
KiCad E.D.A. 9.0.8		Id: 3/3